

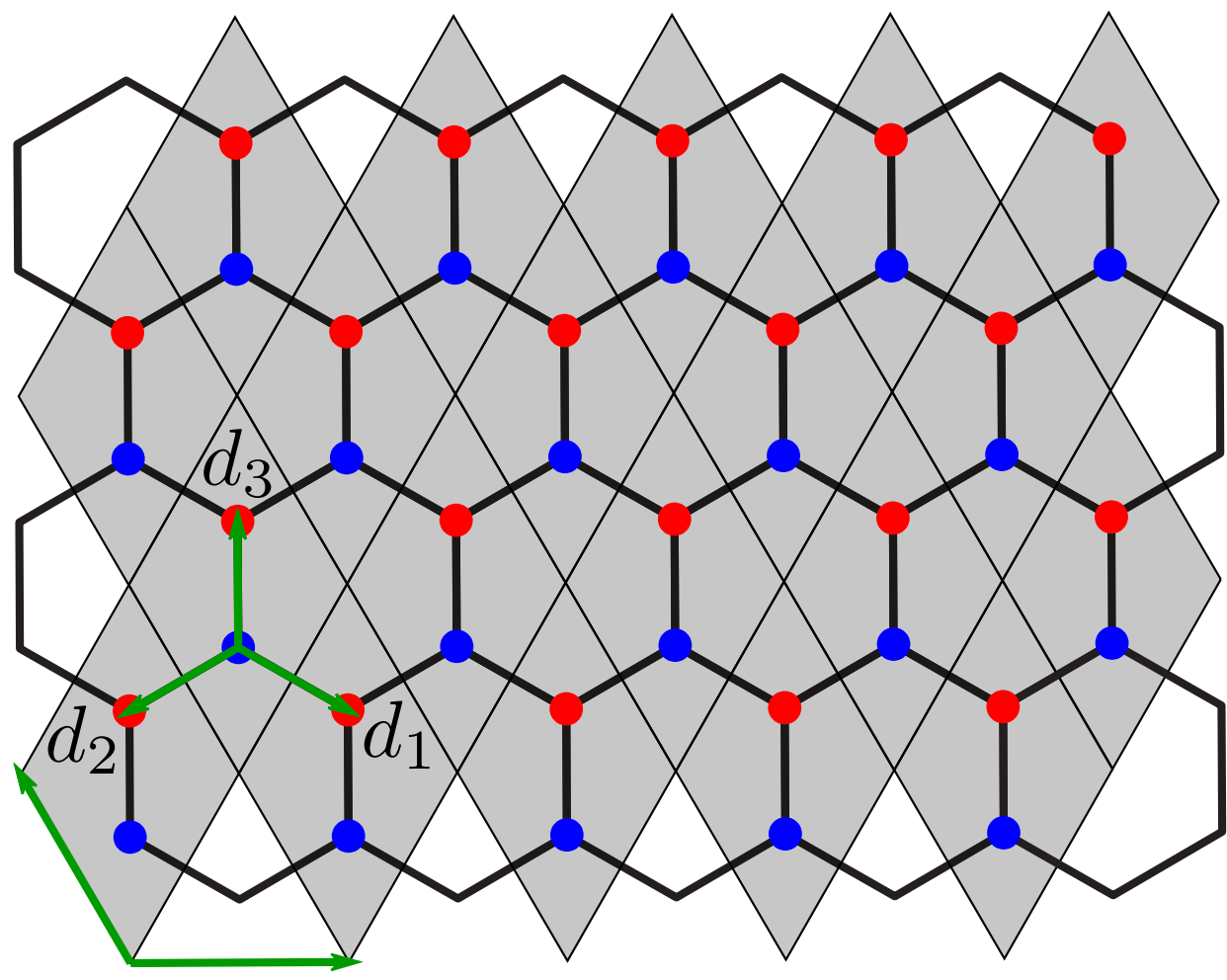
Curvature and Torsion in the Continuum Theory of Lattice Defects

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Offen im Denken

Continuum Theory - Graphene



Tight-binding Hamiltonian

$$H = \sum_{\vec{r}} \sum_{j=1}^3 \left(a_{\vec{r}}^\dagger b_{\vec{r}+\vec{d}_j} + b_{\vec{r}+\vec{d}_j}^\dagger a_{\vec{r}} \right)$$

Fourier transform

$$i\partial_t \vec{\psi}(\vec{k}) = \mathbf{h}(\vec{k}) \vec{\psi}(\vec{k})$$
$$\mathbf{h}(\vec{k}) = \begin{pmatrix} 0 & f(\vec{k}) \\ f^*(\vec{k}) & 0 \end{pmatrix}, \quad f(\vec{k}) = \sum_{j=1}^3 e^{i\vec{k} \cdot \vec{d}_j}$$

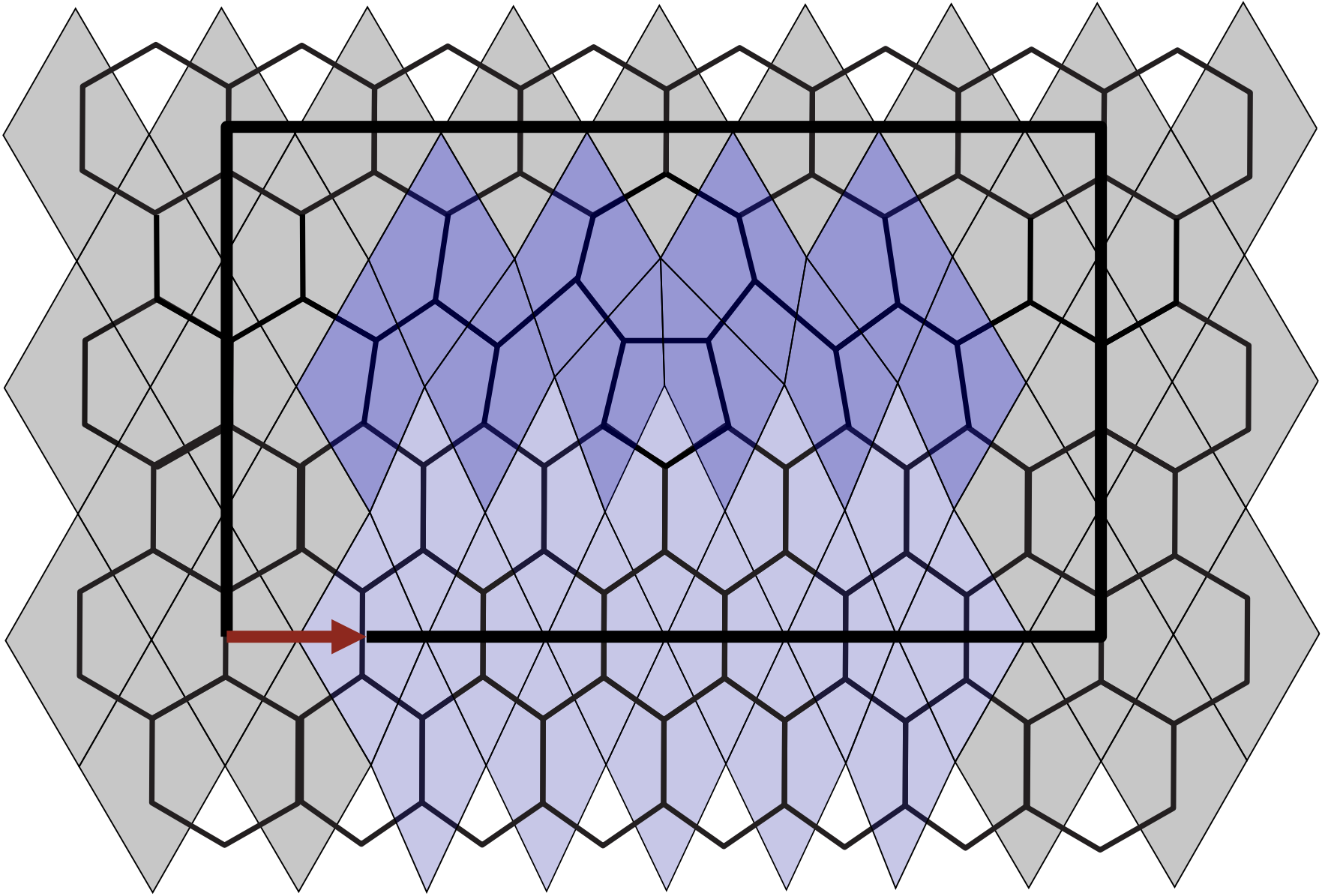
Linear dispersion around Dirac points

$$\vec{k} = \vec{K}^{(s)} + \delta\vec{k} \implies \mathbf{h}(\vec{k}) \approx -\frac{3}{2} \delta\vec{k} \cdot \vec{\sigma}$$

Continuum limit

$$i\partial_t \vec{\phi}(\vec{r}) = i c \vec{\sigma}^i \partial_i \vec{\phi}(\vec{r}), \quad \rho = \vec{\phi}^\dagger \vec{\phi}, \quad \vec{j} = -c \vec{\phi}^\dagger \vec{\sigma} \vec{\phi}$$
$$\Updownarrow$$
$$i\gamma^\mu \partial_\mu \phi = 0, \quad \gamma^\mu = \sigma^\mu \sigma^3, \quad \mu \in \{0, 1, 2\}$$

Burgers vector in 5-7 defects



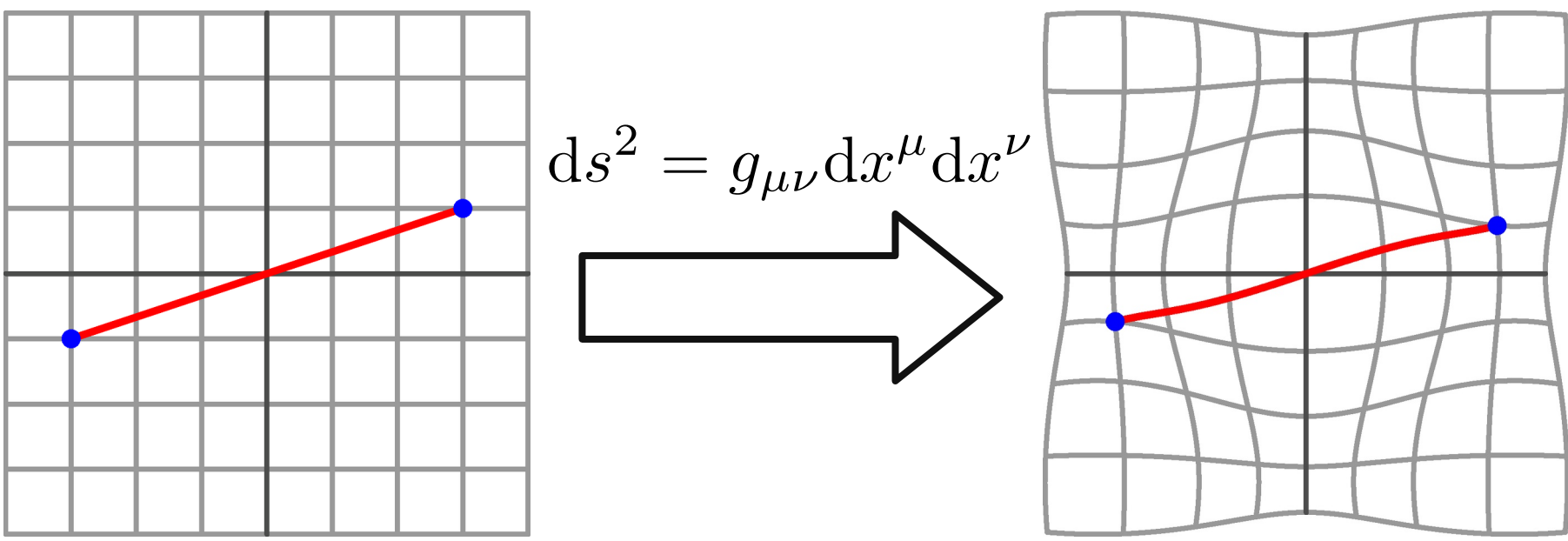
One 5-7 defect causes a lattice dislocation which can be described by the Burgers vector $\vec{b} = \sqrt{3} \vec{u}_x$

Riemannian Geometry

Riemannian Manifold $\mathcal{M}$ equipped with metric $g_{\mu\nu}$ and connection $\Gamma^\lambda_{\mu\nu}$.

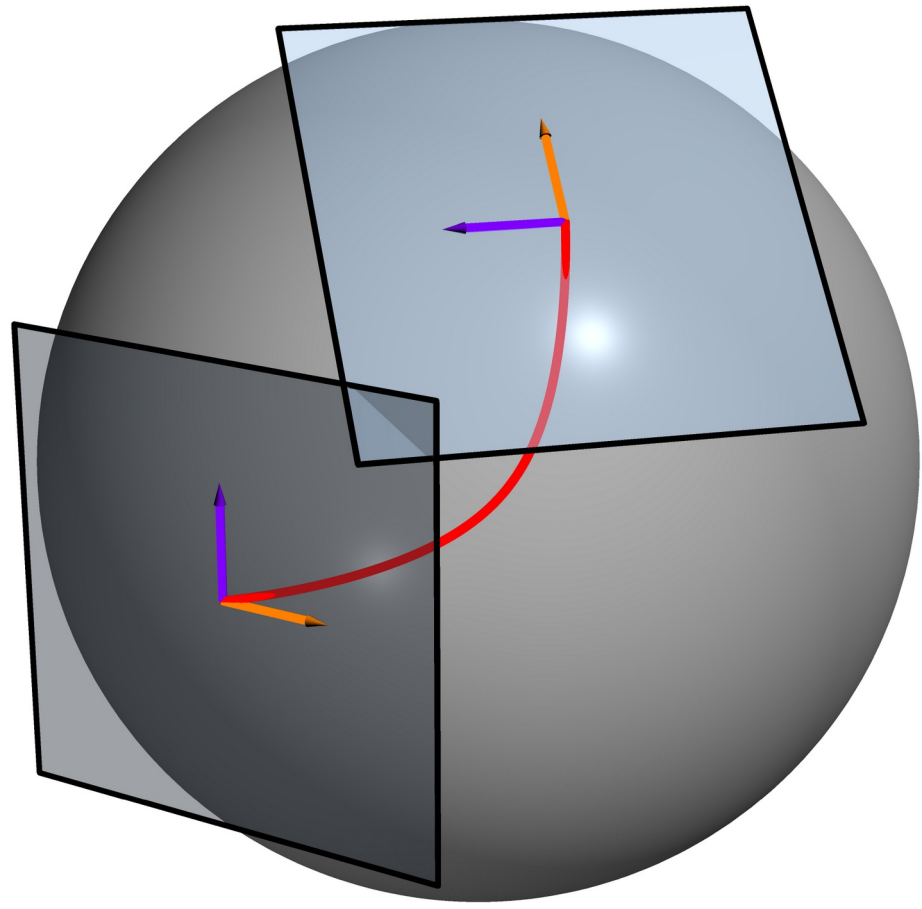
Metric

Defines distances and angles.



Connection

Connects nearby **tangent spaces**, which allows the definition of **vector fields** as functions on the manifold which take on values in the tangent space.



The unique torsion-free, metric compatible connection is called the Levi-Civita connection:

$$\{\lambda_{\mu\nu}\} = \frac{1}{2} g^{\lambda\rho} (\partial_\nu g_{\rho\mu} + \partial_\mu g_{\rho\nu} - \partial_\rho g_{\mu\nu})$$

Covariant derivative

TODO:

- GR basics
- Geodesics/Autoparallels
- Curvature
- Torsion

TODO:

- Numerical implementation of the tight binding hamiltonian
- details on defect modeling
- unitary timeevolution

TODO:

- Numerical results (tight binding vs. autoparallels)
- Plot Scattering angle over defects

TODO:

- In depth visual explanation of torsion and how it is analogous to the burgers vector

TODO:

- Torsion <-> metric mapping
- Riemann curvature
- Autoparallels
- > Problem with parallel transport and tangent space in general

TODO:

- Outlook
- Dirac equation on curved backgroun with torsion
- Numeric simulations of this for some torsion fields

TODO:

- References